
Teacher-Supervised Failure Detection for Distilled Semantic Segmentation under Domain Shift

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Abstract

1 Semantic segmentation enables perception in safety-critical systems, but under
2 distribution shift, these models frequently output dangerous, highly confident
3 yet inaccurate predictions. In these cases, existing confidence measures such as
4 MSP, entropy, MC-Dropout, and MaxLogit struggle to estimate reliability from
5 the student’s output distribution and dangerously degrade on confident failures.
6 We introduce a teacher-supervised, 67K-parameter dense guardrail trained on
7 frozen student logits for distilled semantic segmentation. Unlike ground-truth-
8 supervised confidence heads, our guardrail learns from teacher-derived dense
9 targets. These targets diverge when the distilled student and teacher disagree,
10 providing a structured, low-overhead reliability metric without invoking the teacher.
11 Evaluated across four datasets, teacher-guided supervision surpasses ground-truth
12 supervised guardrails under domain shift, with CF-AUROC improvements of
13 +0.6 to +2.0 pp at $\text{MSP} \geq 0.85$ growing to +5 to +13 pp at stricter MSP thresholds.
14 The largest advantage is seen in high-confidence predictions, where detecting
15 silent failures (confident but incorrect predictions) is most critical: on BDD100K
16 at $\text{MSP} \geq 0.97$, our guardrail improves CF-AUROC by +22 pp over MSP and
17 +13 pp over an equivalent ground-truth supervised head. Our results show that
18 teacher-derived dense supervision is effective at detecting silent failures in distilled
19 segmentation models under distribution shift.

20 1 Introduction

21 Semantic segmentation enables perception in safety-critical systems such as autonomous driving
22 and robotic manipulation, where each pixel prediction propagates into downstream planning and
23 control. Deploying these models on edge hardware typically relies on knowledge distillation (KD) [1],
24 with structured variants for dense prediction [2]. However, distilled students inherit the teacher’s
25 proficiency unevenly. Under domain shift (weather, lighting, camera artifacts), they produce silent
26 confident failures, confident predictions that are wrong, which downstream systems act on without
27 warning. Selective prediction addresses this by abstaining or deferring on unreliable inputs, exchanging
28 coverage for lower prediction risk [3]. However, strong reliability estimations are required for
29 effectiveness. Post-hoc confidence scores (MSP, entropy, MC-Dropout) use the student’s output
30 distribution, which is uninformative when the student is confident. MaxLogit correlates with OOD
31 detection but uses logit magnitudes rather than learned failures, and degrades at stricter confidence
32 thresholds where silent failures are most important. We address this by training a 67K-parameter
33 dense guardrail head on frozen student logits to predict per-pixel failures. The guardrail is supervised
34 by teacher-derived dense targets that diverge on student-teacher disagreement to produce an error
35 signal that adapts to individual inputs rather than being frozen to source-domain labels. Combined
36 with corruption augmentation, the guardrail is a reliable failure score at inference without invoking
37 the teacher at sub-teacher latency. We train on Cityscapes and evaluate on three out-of-distribution

(OOD) benchmarks (ACDC, IDD, BDD100K), comparing teacher-derived supervision to an equivalent ground-truth-supervised baseline.

Contributions.

1. The guardrail: a teacher-supervised failure detector for distilled semantic segmentation that flags silent confident failures under domain shift, trained with teacher-derived dense targets and corruption augmentation.
2. Evidence that teacher-derived dense supervision transfers better under shift than ground-truth supervision, with the largest advantage at strict confidence thresholds where silent failures are most dangerous and post-hoc baselines collapse toward random.
3. A cost-reliability analysis showing the teacher heads’ advantage over matched GT-supervised heads increases at strict confidence thresholds with competitive or improved performance relative to post-hoc baselines in the most difficult cases, at a favorable cost-reliability trade-off (§3.5).

2 Method

Overview. We study image-level failure detection for distilled semantic segmentation under domain shift. A strong teacher model T is first used to distill a lightweight student S . After distillation, both networks are frozen. We then train the guardrail G on top of the student’s logits to predict failure risk. The central question in this work is not the architecture of G , but the supervision signal used to train it. To isolate that question, we compare two cases: one supervised by teacher-derived dense targets and the other supervised by ground-truth-derived dense targets, while fixing the guardrail architecture, optimizations, and augmentations.

Notation. For an input image x , let

$$z^S(x), z^T(x) \in \mathbb{R}^{H \times W \times C}$$

denote the student and teacher logit tensors, where $H \times W$ is the output spatial resolution and C is the number of semantic classes. Let

$$\hat{y}_{ij}^S = \arg \max_c z_{ijc}^S, \quad \hat{y}_{ij}^T = \arg \max_c z_{ijc}^T$$

be the student and teacher predictions at pixel (i, j) , let y_{ij} denote the ground-truth label, and let $m_{ij} \in \{0, 1\}$ be a valid-pixel mask that excludes ignored labels. We write

$$\text{CE}(z, y) = -\log(\text{softmax}(z)_y)$$

for pixelwise cross-entropy.

Student–teacher setup. The teacher T is a SegFormer-B5 model (84.7M parameters), and the student S is a SegFormer b0/b1/b2 model [4] (3.7M–24.7M parameters). We train in two stages. First, S is pretrained on Cityscapes with standard semantic segmentation supervision. Next, S is distilled from a frozen teacher with a combination of segmentation loss, logit distillation with KL divergence (temperature $\tau = 2$), and a pairwise-affinity feature loss inspired by structured distillation for dense prediction [2]. After this stage, both T and S are fixed. All subsequent learning is on the guardrail head G and operates on the detached student logits. Gradients are never propagated into T or S during guardrail training.

Guardrail architecture. The guardrail G is a lightweight convolutional head with the input of the student’s per-pixel logits. It consists of three 3×3 convolutional layers with channel dimensions

$$19 \rightarrow 64 \rightarrow 64 \rightarrow 32,$$

and two task-specific output heads. The first head predicts a dense disagreement map

$$\hat{d} \in [0, 1]^{H \times W},$$

and the second predicts a dense, continuous risk map

$$\hat{r} \in \mathbb{R}^{H \times W}.$$

The full module contains 66,819 parameters. In the multi-task configuration, both outputs are generated by a shared convolutional encoder and jointly optimized.

Supervision. We compare two supervision cases for training G , which differ only in how the dense targets are constructed. In both cases, the guardrail only receives the detached student logits z^S and predicts the pair $(\hat{d}, \hat{r})$. The architecture, optimizer, corruption policy, and inference rule are otherwise identical.

In the teacher-supervised regime, the binary target indicates teacher–student disagreement at each pixel:

$$d_{ij}^T = \mathbf{1}[\hat{y}_{ij}^S \neq \hat{y}_{ij}^T].$$

The continuous target measures the student’s excess loss relative to the teacher in the teacher’s preferred class:

$$r_{ij}^T = \text{CE}(z_{ij}^S, \hat{y}_{ij}^T) - \text{CE}(z_{ij}^T, \hat{y}_{ij}^T).$$

In the ground-truth-supervised regime, the binary target indicates whether the student’s prediction is incorrect compared to the labeled class:

$$d_{ij}^{\text{GT}} = \mathbf{1}[\hat{y}_{ij}^S \neq y_{ij}],$$

and the continuous target is the student’s cross-entropy with respect to the ground-truth label:

$$r_{ij}^{\text{GT}} = \text{CE}(z_{ij}^S, y_{ij}).$$

This gives two controlled training classes that differ only in the source of supervision. The resulting comparison isolates whether teacher-derived dense risk targets transfer better under domain shift than dense targets derived from source-domain correctness.

Training objective. The disagreement head is trained with masked binary cross-entropy, and the risk head is trained with masked smooth- ℓ_1 loss, both averaged over valid pixels only. The full guardrail objective is

$$\mathcal{L}_G = \lambda_{\text{dis}} \mathcal{L}_{\text{BCE}}(\hat{d}, d) + \lambda_{\text{risk}} \mathcal{L}_{\text{S}\ell_1}(\hat{r}, r),$$

where (d, r) denotes either the teacher-supervised targets (d^T, r^T) or the ground-truth-supervised targets $(d^{\text{GT}}, r^{\text{GT}})$. Single-task variants are obtained by setting one loss weight to zero; the multi-task model optimizes both terms jointly.

Inference. At test time, we use the guardrail as a dense risk predictor rather than as a teacher simulator. Given an image x , the guardrail produces a per-pixel risk map $\hat{r}(x)$, which gets combined into a scalar image-level failure score.

$$g(x) = \sigma\left(\frac{\sum_{ij} m_{ij} \hat{r}_{ij}(x)}{\sum_{ij} m_{ij}}\right) \in [0, 1],$$

where $\sigma(\cdot)$ is the sigmoid function. Larger values of $g(x)$ indicate a higher predicted failure risk. The disagreement head is only used as an auxiliary training signal to improve the shared representation, so the default inference-time score in the main experiments is computed only from the risk head. A threshold on $g(x)$ can flag high-risk inputs or trigger teacher deferral. Because $g(x)$ depends only on the frozen student logits and the lightweight guardrail head, inference requires a single forward pass of S and G and no teacher invocation.

Training procedure. To broaden the supervision signal beyond clean source-domain inputs, we apply online corruption augmentation during guardrail training. With probability $p = 0.5$, each training image is perturbed by a sampled corruption drawn from a set including fog, blur, noise, and underexposure. Whenever a corruption is applied, both the teacher and the student are re-evaluated on the perturbed input, so that the dense supervision targets match the input seen by G . This exposes the guardrail to a range of shifted patterns during training while preserving consistency between inputs and supervision.

3 Experiments

Datasets. Cityscapes-val (500 images, in-domain) [5], ACDC-val (406 images; fog/night/rain/snow) [6], IDD-val (978 images; India, semantic shift) [7], BDD100K-val (1000 images; US geographic shift) [8]. All models trained on Cityscapes only.

Table 1: CF-AUROC @ $\text{MSP} \geq 0.85$ (mit-b1). GT heads use the same architecture and training as teacher heads. Only the supervision source differs.

	MSP	Ent	MaxL	MC-D	GT-Dis	GT-Gap	T-Multi	Δ_{GT}
ACDC	.587	.592	.596	.593	.589	.599	.604	+0.6
IDD	.573	.578	.543	.577	.575	.592	.603	+1.2
BDD	.657	.680	.732	.682	.738	.716	.758	+2.0

Baselines. We compare against MSP [9], temperature-scaled MSP [10], MC-Dropout [11], MaxLogit [12], and entropy. MaxLogit uses the maximum logit value as the confidence score for OOD detection. Multi-task is the default unless stated otherwise. Figures report $\pm 1\sigma$ over $N = 3$ training seeds.

Metrics. F-AUROC, CF-AUROC at MSP threshold $\tau \in \{0.85, 0.95, 0.97\}$.

3.1 Teacher vs. ground-truth supervision

Table 1 shows the main result: the teacher-supervised multi-task head (T-Multi) outperforms both GT-supervised heads (GT-Dis, GT-Gap) on all three OOD datasets. The Δ_{GT} column reports the gap between T-Multi and the best GT head.

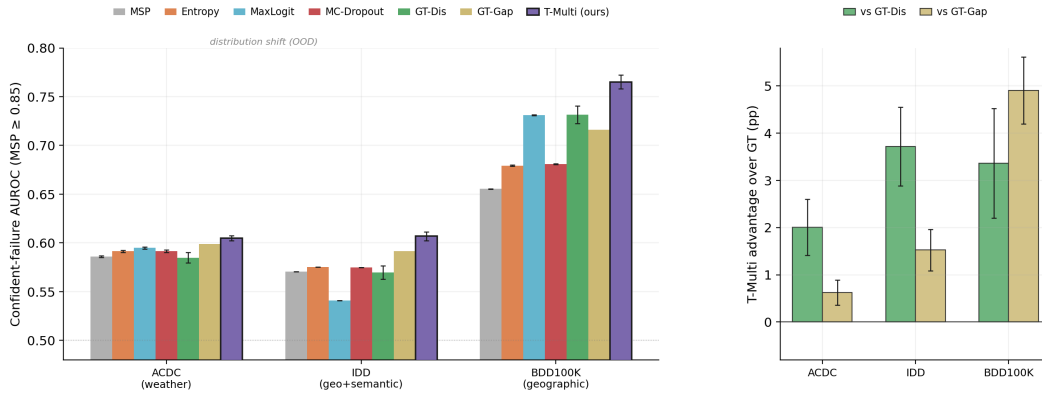


Figure 1: Left: CF-AUROC @ $\text{MSP} \geq 0.85$ across methods and datasets (mit-b1). Right: T-Multi improvement over each GT baseline in percentage points.

Why teacher supervision transfers better. Ground-truth labels define “correctness” relative to the source domain’s labels. Under domain shift, the distribution of student errors changes, so the student makes different kinds of mistakes on fog vs. rain vs. geographic shift. The GT labels, by contrast, are frozen to source-domain patterns. The teacher is re-evaluated on each training image (including corrupted variants), offering an error signal that adapts to the input. Under domain shift, teacher-derived dense targets transfer better than dense targets defined only by source-domain correctness.

Table 2: CF-AUROC at strict MSP thresholds (mit-b1). At $\text{MSP} \geq 0.95$, all post-hoc scores converge towards random. T-Multi maintains discrimination, with a growing advantage over GT. ACDC at $\text{MSP} \geq 0.97$ has $n = 23$, so results have not been included.

	τ	MSP	MaxL	MC-D	GT-Dis	GT-Gap	T-Multi
ACDC	0.85	.587	.596	.593	.589	.599	.604
	0.95	.437	.534	.432	.501	.508	.563
IDD	0.85	.573	.543	.577	.575	.592	.603
	0.95	.501	.505	.509	.506	.519	.541
	0.97	.485	.557	.499	.492	.513	.502
BDD	0.85	.657	.732	.682	.738	.716	.758
	0.95	.487	.589	.530	.608	.573	.662
	0.97	.341	.380	.396	.439	.431	.566

The T-Multi advantage over GT grows at stricter thresholds. At $\text{MSP} \geq 0.95$ on ACDC, T-Multi 0.563 vs. GT-Gap 0.508 (+5.5 pp). On BDD at 0.95, T-Multi 0.662 vs. GT-Dis 0.608 (+5.4 pp). At 0.97 on BDD, T-Multi maintains 0.566 while GT-Dis drops to 0.439 (+12.7 pp). The widening gap suggests the teacher signal provides more robust features in the hardest cases. MaxLogit’s AUROC declines steeply with tighter MSP thresholds as softmax saturation compresses the confident subset’s logit-magnitude spread.

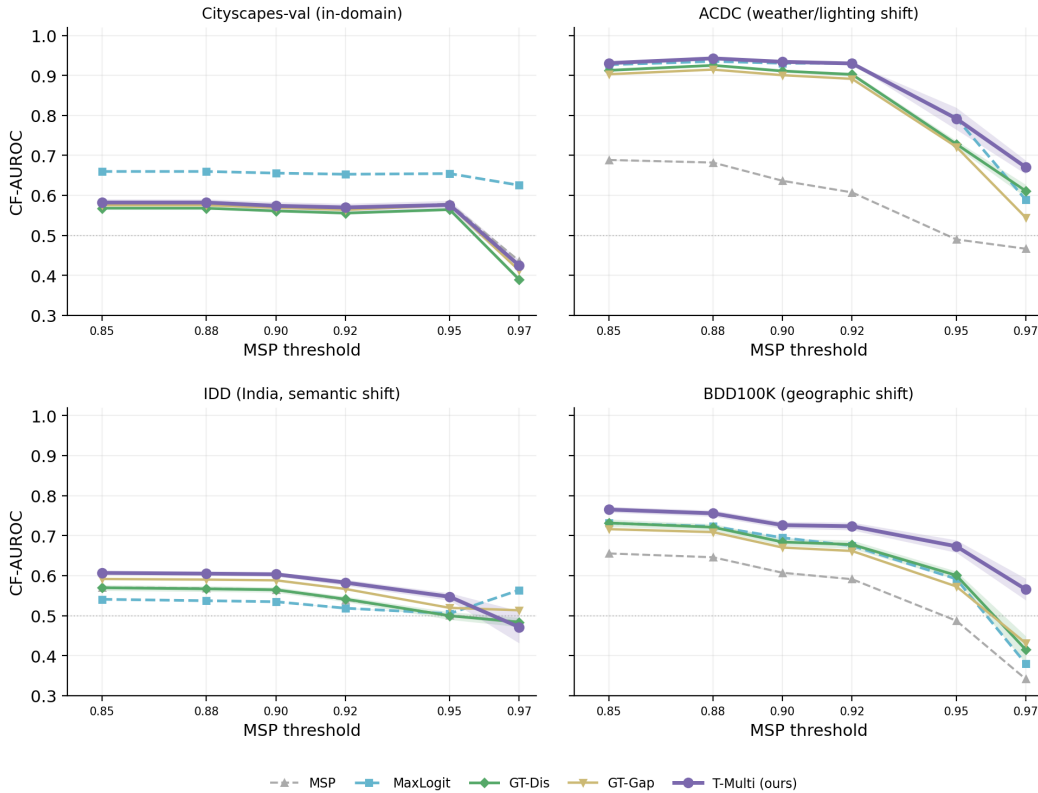


Figure 2: CF-AUROC across MSP thresholds (mit-b1). T-Multi degrades most slowly on ACDC and BDD.

3.3 Within-condition analysis

Table 3: Within-condition failure AUROC (ACDC, mit-b1). Bottom rows: macro-average (mean of per-condition AUROCs), condition-only baseline (assign each image its condition’s mean risk), and pooled. The gap between pooled and macro-averaged results shows that much of the pooled gain comes from separating conditions rather than from uniformly stronger failure rankings within each condition.

Cond.	MSP	MaxL	MC-D	GT-Dis	GT-Gap	T-Multi
Fog	.495	.576	.503	.519	.521	.600
Night	.607	.760	.625	.670	.659	.590
Rain	.734	.628	.749	.747	.762	.764
Snow	.552	.664	.554	.602	.628	.636
Macro-avg	.597	.657	.608	.634	.643	.648
Condition-only	0.816 (assigns condition mean risk)					
Pooled	.692	.927	.710	.910	.903	.928

Condition-only baseline. A score that assigns each image its condition’s mean student risk achieves 0.816 pooled F-AUROC. T-Multi’s pooled 0.928 exceeds this by +11.2 pp, demonstrating it identifies signal beyond the condition class.

Macro-average. Averaging across conditions removes the between-condition signal entirely. MaxLogit leads (0.657), followed by T-Multi (0.648) and GT-Gap (0.643). T-Multi beats MSP (+5.1 pp) and MC-Dropout (+4.0 pp) within-condition, but MaxLogit’s within-condition ranking is stronger by 0.9 pp. T-Multi’s pooled advantage over MaxLogit thus comes mainly from the condition-level signal rather than learned patterns.

Night weakness. Night images have uniformly high student risk (mean 0.82, std 0.05; nearly all are failures), leaving almost no within-condition variance to detect. The guardrail score is near-constant within night (std = 0.07, corr = 0.25 with student risk), while MaxLogit’s logit magnitudes vary more (std = 0.58, |corr| = 0.43).

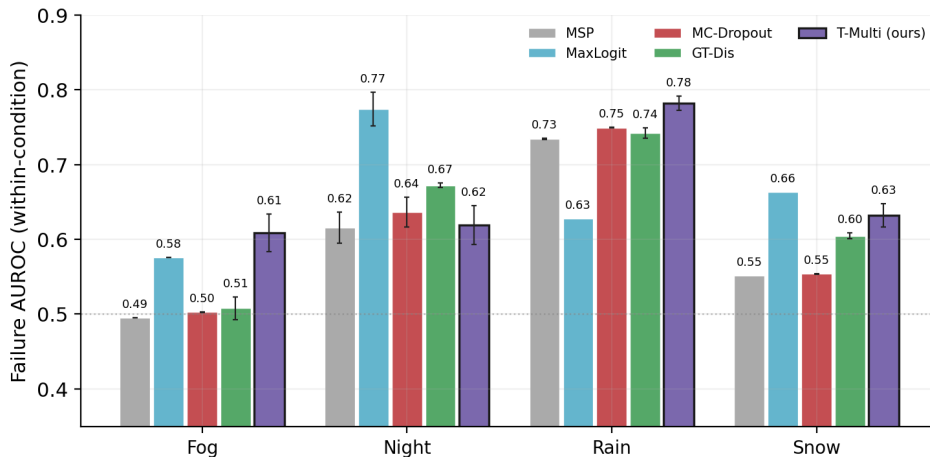


Figure 3: Within-condition failure AUROC on ACDC (mit-b1). T-Multi wins fog and rain; MaxLogit dominates night and snow.

3.4 Supervision mode ablation

Table 4: Full ablation of supervision modes (b1, CF-AUROC @ $\text{MSP} \geq 0.85$). Teacher-supervised modes are strongest on BDD, while ACDC remains mixed across target types.

	MSP	Scalar	GT-Dis	GT-Gap	T-Dis	T-Gap	T-Multi
City	.579	.613	.566	.576	.574	.583	.571
ACDC	.587	.569	.589	.599	.582	.597	.604
IDD	.573	.570	.575	.592	.585	.617	.603
BDD	.657	.754	.738	.716	.745	.747	.758

The ablation separates the two design axes for the guardrail head: the supervision source (teacher vs. GT) and the target shape (dense disagreement, dense gap, or both). Along the source axis, the teacher signal’s largest benefit is on BDD, where every teacher mode beats every GT variant (best teacher 0.758 vs. best GT 0.738, +2.0 pp); it narrows on IDD (+2.5 pp for T-Gap over GT-Gap) and is marginal on ACDC (+0.5 pp for T-Multi), consistent with the per-condition analysis that shows the ACDC gains are primarily from learned condition rather than within-condition rankings. Along the shape axis, a continuous gap target consistently outperforms binary disagreement: T-Dis is inferior to T-Gap and T-Multi on every OOD dataset and even loses to both GT variants on ACDC (0.582 vs. 0.589/0.599). T-Multi combines both axes and is the default.

Scalar baseline as negative control. The scalar row trains the same architecture on a single image-level target (student_risk—teacher_risk) instead of dense per-pixel supervision. Under the joint evaluation, results degrade to near-MSP performance on ACDC (0.569) and IDD (0.570), confirming that per-pixel supervision turns the teacher signal into a usable score. The image-level benefit target itself is structurally unlearnable ($R^2 \leq 0.06$ cross-validated). Its BDD result (0.754) comes from student-risk features the head picks up incidentally rather than from predicting its stated target. By contrast, the dense teacher-supervised modes retain a stronger signal in the high-confidence regime (Table 2).

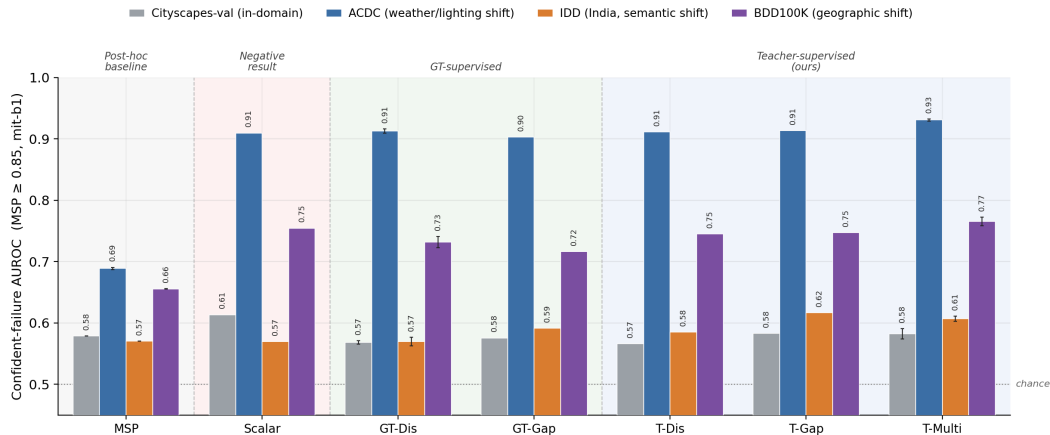


Figure 4: Full supervision ablation (mit-b1, CF-AUROC @ $\text{MSP} \geq 0.85$). Teacher-supervised modes are strongest on BDD and mixed on ACDC; T-Multi is the best overall teacher-supervised variant.

3.5 Inference cost and reliability comparison

Failure detection is only practical when the inference cost is justified by the reliability gain. Table 5 reports per-image inference cost alongside CF-AUROC at $\text{MSP} \geq 0.95$, the high-confidence regime where post-hoc scores are weakest and silent failures are most dangerous.

Pareto analysis. The guardrail improves over MC-Dropout, with +9.9 pp higher mean AUROC at 33% lower latency despite MC-Dropout requiring four stochastic forward passes. Against MaxLogit,

Table 5: Per-image inference latency vs. confident-failure AUROC at $\text{MSP} \geq 0.95$ on three datasets (mit-b1, $1 \times \text{NVIDIA L40S}$, batch size=4). ACDC is the per-condition mean over fog/night/rain/snow. Latencies are reported as the median over 500 Cityscapes-val images after 50 warm-up passes, with CUDA synchronization on each forward pass.

Method	Latency (ms)	ACDC	IDD	BDD100K	Mean OOD
Student (MSP)	4.9	.437	.501	.487	.475
MaxLogit	4.9	.534	.505	.589	.543
MC-Dropout (4 passes)	25.2	.432	.509	.530	.490
Student + guardrail (T-Multi, ours)	16.8	.563	.541	.662	.589
Teacher (deferral oracle)	26.1	1.00	1.00	1.00	1.000

the tradeoff is +11.9 ms of latency for +4.6 pp mean AUROC, widening to +7.3 pp on BDD100K. Against the deferral oracle, the guardrail closes 22% of the MSP-to-teacher gap on average (34% on BDD) without ever invoking the teacher.

Cost decomposition. The guardrail head adds 11.9 ms to the student’s 4.9 ms forward pass, for a total of 16.8 ms. This overhead is independent of backbone size as the head uses the segmentation output resolution. MC-Dropout’s 25.2 ms uses four stochastic student passes, and the teacher’s 26.1 ms is a single 84.7M-parameter forward pass. The guardrail therefore offers a fixed cost while beating both alternatives in latency.

Selective deferral. The guardrail’s risk score gates teacher invocation. If the teacher is invoked on flagged inputs at flag rate f , the expected per-frame latency is

$$\mathbb{E}[L] = L_{\text{student}} + L_{\text{guardrail}} + f \cdot L_{\text{teacher}}.$$

At $f = 0.1$, $\mathbb{E}[L] = 19.4$ ms: 15% above guardrail-alone latency and 26% below full teacher inference. Because inputs are only flagged on real failures (§3.3), this method approaches teacher-level reliability while keeping latency close to the guardrail.

4 Discussion

Our results support two main conclusions. First, learned dense failure predictors are stronger than standard post-hoc confidence metrics under distribution shift. Second, within this class of methods, teacher-guided supervision consistently outperforms matched GT supervision. This gain is modest at standard operation, but increases in the high-confidence regime, where silent failures are most dangerous. Simultaneously, the comparison to post-hoc baselines is not uniform: MaxLogit is a strong baseline, especially on structured weather shift, while the teacher-guided head degrades more gracefully on ACDC and BDD as confidence thresholds tighten. Together, these results suggest that the proposed method is most useful when failures are not well captured by a single global confidence metric and require learned spatial structure.

The ablations show that most of the improvement over MSP comes from learning a dense predictor on top of frozen student logits with corruption-aware training. Teacher supervision also provides a smaller but consistent benefit over GT supervision under shift. This also helps position the method relative to ConfidNet-style approaches [13]: our GT-Dis baseline is the matched GT-supervised analog, so the teacher-versus-GT comparison isolates the value of the teacher signal rather than changes in architecture or optimization. Overall, teacher supervision improves transfer once a strong, dense failure-prediction architecture is in place.

The method also has limitations. The within-condition analysis shows that the method is not uniformly superior: night scenes are difficult, and in-domain performance has little to no advantage over confidence baselines. In addition, the method does not benefit from teacher inference at test time, although the risk can trigger teacher deferral. The conclusion, supported by the results, is that teacher-guided dense supervision improves failure detection under domain shift, with the clearest benefit in the high-confidence regime, where post-hoc confidence is least reliable.

Broader impacts. This work estimates risk for semantic segmentation under domain shift, with a potential positive impact in safety-critical perception settings (such as driving) by detecting high

confidence failures before downstream decisions are made using them. However, the guardrail is not a safety certificate: false negatives could still allow unsafe predictions and false positives may unnecessarily defer and degrade system performance. The experiments use driving-scene datasets, so deployment could also learn dataset biases across geography, weather, lighting, and road infrastructure. Any real-world use should treat the guardrail as a component of a larger safety stack rather than a standalone guarantee of safety.

5 Related work

Selective prediction uses model confidence to abstain from predicting. MSP [9], temperature scaling [10], and MC-Dropout [11] are standard post-hoc scores. MaxLogit [12] uses raw logit magnitude for OOD detection. ConfidNet trains an auxiliary head to predict True Class Probability (TCP) from ground-truth labels [13]. Our method replaces ground-truth-derived dense targets with teacher-derived dense targets, which transfer better under shift because they adapt to the teacher’s response.

Uncertainty distillation. Deep Ensembles transfer uncertainty from ensemble teachers, typically requiring 3–5 independently trained models [14]. DUDES distills ensemble uncertainty into a single-pass segmentation model [15]. Our method is complementary and only uses a single teacher and a 67K-parameter head rather than distilling ensemble uncertainty.

Knowledge distillation for segmentation. SegFormer and related architectures are compressed via standard KD [1, 4]. Structured KD extends this with pixel-wise, pairwise, and holistic dense-prediction objectives [2]. Prior work focuses on closing the mIoU gap. Our work complements this by equipping the distilled student with a post-distillation reliability measure.

6 Conclusion

Teacher-supervised dense targets provide a stronger signal for failure detection under distribution shift than dense targets derived from ground-truth correctness. A 67K-parameter head trained with these targets outperforms identically architected GT-supervised heads under domain shift (+0.6 to +2.0 pp at standard thresholds, widening to +5 to +13 pp at strict ones on ACDC and BDD), while in-domain behavior remains mixed. Compared to post-hoc confidence baselines, the guardrail remains competitive overall, stays close to MaxLogit on structured weather shift, and exceeds it on diffuse geographic shift (BDD, +2.6 pp CF-AUROC). The guardrail additionally improves the cost-reliability trade-off, Pareto-dominating MC-Dropout and closing 22% of the MSP-teacher gap on out-of-distribution data without ever invoking the teacher (§3.5).

The main driver of improvement over post-hoc methods is the guardrail architecture with corruption augmentation, which benefits all learned heads, including GT-supervised ones. The teacher signal adds a secondary but consistent gain. The guardrail is most useful when failure patterns vary across images (ACDC, BDD), and offers less additional benefit when that variation is limited (IDD, Cityscapes).

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293 A Silent confident failure comparison

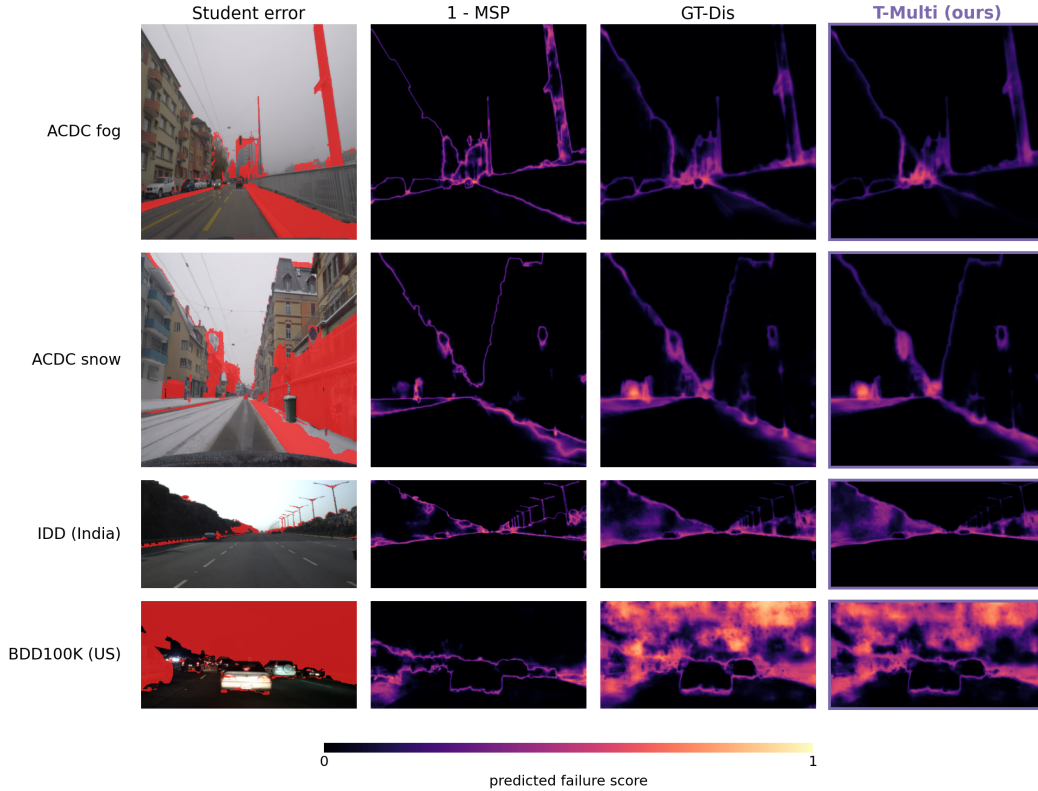


Figure 5: Qualitative failure detection across four out-of-distribution scenarios (mit-b1). Student pixel error overlaid in red on the input image. Across all four scenarios, 1–MSP activates almost exclusively along object boundaries and misses the majority interior regions where the student fails. The learned heads recover the boundary structure and additionally fire on failed interiors, with T-Multi showing the brightest foreground response on ACDC fog and BDD100K, where silent confident failures are most spatially concentrated.

294 B Selective prediction

Table 6: AUGRC (mit-b1). Images ranked by risk.

	MSP	MaxL	TempMSP	MC-D	GT-Dis	GT-Gap	T-Multi
City	.422	.415	.422	.422	.423	.422	.422
ACDC	.634	.601	.625	.630	.610	.612	.610
IDD	.633	.636	.635	.632	.634	.632	.632
BDD	.533	.529	.531	.529	.521	.524	.515

295 AUGRC is a threshold-free metric that measures the area under the risk-coverage curve. On BDD,
 296 T-Multi (0.515) beats MaxLogit (0.529) and GT-Dis (0.521). On ACDC, T-Multi ties GT-Dis (both
 297 0.610). MaxLogit achieves the best AUGRC (0.601) on this dataset, due to the condition-level
 298 separation in logit magnitudes. The learned heads have no advantage over post-hoc methods for
 299 in-domain images.

300 C Risk-coverage curves

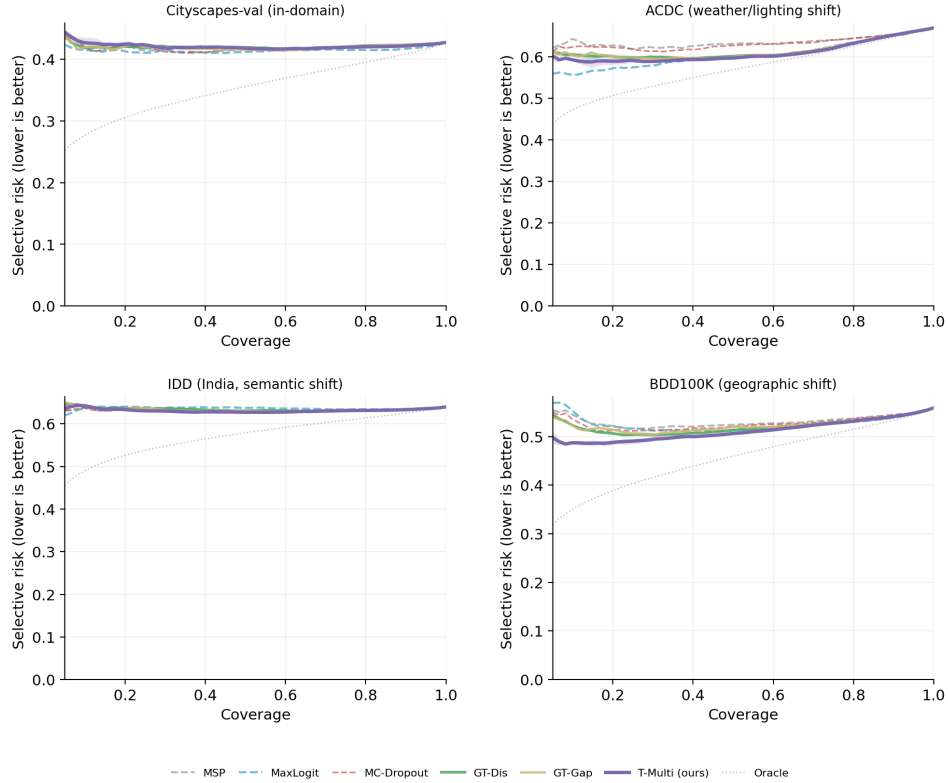


Figure 6: Risk-coverage curves (mit-b1). T-Multi is competitive with GT baselines on ACDC and achieves the lowest selective risk on BDD.

301 D Assets and licenses

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